

DATA INSIGHTS, RECOMMENDATIONS AND IMPLEMENTATION

Agricultural Value Chain Loss & Profitability AnalysisProject Overview

Post-harvest losses and inefficiencies in agricultural supply chains significantly reduce farmer income and overall productivity. In this project, I analyzed production, logistics, and sales data from an AgriTech startup to identify where losses occur and uncover opportunities to improve profitability across crops, regions, and farmers.

Business Problem

The startup aims to answer key questions:

1. Which crops generate the highest profit margins by region?
2. Which regions experience the most transportation losses?
3. What is the average logistics cost per kg sold?
4. How much revenue is lost due to spoilage or inefficient transport?
5. Which farmer groups or markets are most efficient?

INSIGHTS

1. High Revenue Loss Across the Value Chain

Approximately **35.6% of total revenue** is lost, representing a significant inefficiency. This level of loss substantially reduces overall profitability. Addressing it is critical for sustainable growth.

2. Spoilage is the Main Driver of Losses

About **92.99% of total revenue loss (₦429.8M)** is due to spoilage. This far outweighs transport-related losses. Improving post-harvest handling and storage can deliver the highest impact.

3. Losses are Concentrated in Central and West Regions

The **Central (₦102M)** and **West (₦101M)** regions experience the highest losses. These regions account for the largest share of inefficiencies. Targeted interventions here can significantly reduce total losses.

4. Transport Losses are Minimal Compared to Spoilage

Transport losses account for only **7.01% (₦32.4M)** of total losses. Loss rates remain low across regions (**~0.6%–0.8%**). This confirms transport is not the primary issue.

5. Storage Capacity is Not the Core Issue

Storage utilization averages around **83% across all regions**, indicating capacity is sufficient. Despite this, losses remain high, pointing to inefficiencies in storage quality or handling. Improving storage practices is key.

6. Strong Profitability with Opportunities to Scale Efficiency

Profit margins are consistently high (~86%+ across crops and regions), with **Sorghum leading**. Top farmers achieve even higher margins (**above 90%**). Scaling these best practices can further boost performance.

Executive Summary

The analysis reveals that approximately **35.6% of total revenue is lost** across the agricultural value chain, with **spoilage accounting for 92.99% (₦429.8M)** of these losses, far exceeding transport-related losses (**7.01%, ₦32.4M**). Losses are heavily concentrated in the **Central (₦102M)** and **West (₦101M)** regions, indicating localized inefficiencies. Despite relatively high **storage utilization (~83%)**, losses remain significant, suggesting that the issue lies in **storage quality and post-harvest handling rather than capacity**. On the performance side, operations are highly profitable, with **profit margins exceeding 86% across crops and regions**, led by Sorghum, while top farmers achieve margins above 90%. This highlights a strong opportunity to **reduce spoilage and scale existing best practices to improve overall efficiency and profitability**.

RECOMMENDATIONS

1. High Revenue Loss (35.6%)

Recommendation:

- Prioritize loss-reduction strategies across the value chain
- Set a target to reduce losses by 10–15% in the short term
- Track loss KPIs monthly to monitor improvements

2. Spoilage dominates losses (92.99%, ₦429.8M)

Recommendation:

- Invest in improved storage infrastructure (cold storage, ventilation)
- Train farmers on post-harvest handling techniques
- Improve market access to reduce storage duration

3. Losses concentrated in Central (₦102M) & West (₦101M)

Recommendation:

- Implement pilot intervention programs in these regions
- Improve road infrastructure and logistics coordination
- Focus resource allocation where impact will be highest

4. Transport losses are low (7.01%, ~0.6–0.8%)

Recommendation:

- Maintain current transport systems
- Only implement targeted improvements in high-loss zones (e.g., West)

- Avoid over-investing in transport at the expense of storage

5. Storage utilization high (~83%) but losses persist

Recommendation:

- Focus on storage efficiency, not expansion
- Introduce better inventory management systems
- Reduce storage time through better demand forecasting

6. High profitability (~86%+) & top farmers (>90%)

Recommendation:

- Identify and document best practices from top-performing farmers
- Scale through training programs and knowledge sharing
- Encourage production of high-margin crops like Sorghum

IMPLEMENTATION PLAN

The implementation plan is divided into short-term actions for immediate impact and long-term investments to address structural inefficiencies.

Action	Timeline	Key Stakeholders	Key Steps	Expected Impact
Establish a loss monitoring system	0–3 Months	Data Team, Operations Team	Define loss KPIs, build tracking dashboard, review losses regularly	Improved visibility into losses and faster decision-making
Improve post-harvest handling practices	0–3 Months	Extension Officers, Farmer Cooperatives	Train farmers on drying, sorting, packaging, and proper handling techniques	Reduction in spoilage losses and improved product quality
Implement targeted interventions in Central & West regions	0–3 Months	Regional Managers, Logistics Team	Deploy pilot programs and monitor loss trends in high-loss regions	Significant reduction in regional revenue losses
Optimize transport operations in high-risk zones	3–6 Months	Logistics Team, Transport Partners	Improve routing and scheduling in regions with higher transport losses	Sustained low transport loss rates and improved efficiency
Improve storage efficiency and inventory management	3–6 Months	Operations Team, IT Team	Introduce inventory tracking and reduce storage duration	Reduced spoilage and better stock management
Scale best practices from top-performing farmers	6–12 Months	Data Team, Field Officers, Trainers	Identify top farmers and replicate their methods across the	Increased profitability and operational

			network	efficiency
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Conclusion

The analysis reveals that while strong profitability opportunities exist—particularly with Sorghum—significant revenue losses persist due to inefficiencies in logistics and post-harvest handling. Addressing these gaps through targeted short-term actions and long-term infrastructure investments can substantially improve value chain efficiency, farmer productivity, and overall profitability.